

Assignment-6

1.Find the sum of array of elements.

```
import java.util.Scanner;
```

```
public class ArraySum {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the number of elements: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        int sum = 0;
```

```
        System.out.println("Enter " + n + " elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
            sum = sum + arr[i];
```

```
        }
```

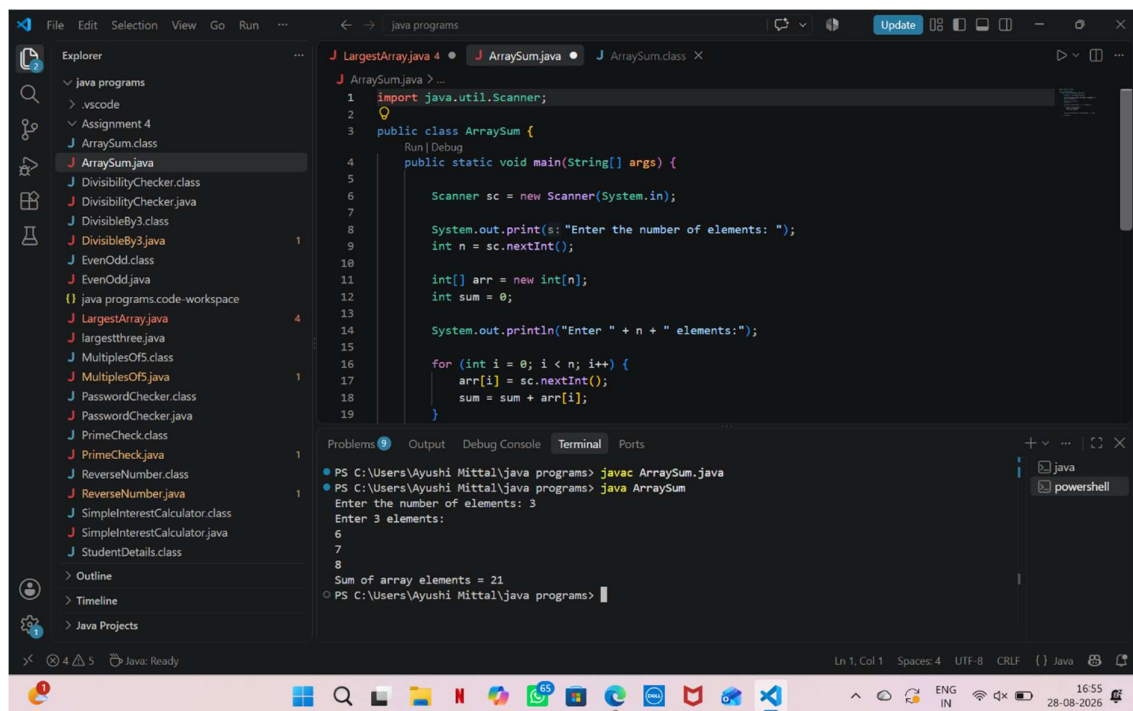
```
System.out.println("Sum of array elements = " + sum);
```

```
sc.close();
```

```
}
```

```
}
```

Output



```
File Edit Selection View Go Run ... java programs Update
Explorer
  java programs
  .vscode
  Assignment 4
  ArraySum.class
  ArraySum.java
  DivisibilityChecker.class
  DivisibilityChecker.java
  DivisibleBy3.class
  DivisibleBy3.java
  EvenOdd.class
  EvenOdd.java
  java programs.code-workspace
  LargestArray.java
  largestthree.java
  MultiplesOf5.class
  MultiplesOf5.java
  PasswordChecker.class
  PasswordChecker.java
  PrimeCheck.class
  PrimeCheck.java
  ReverseNumber.class
  ReverseNumber.java
  SimpleInterestCalculator.class
  SimpleInterestCalculator.java
  StudentDetails.class
  > Outline
  > Timeline
  > Java Projects
J LargestArray.java 4 • J ArraySum.java • J ArraySum.class X
J ArraySum.java > ...
1 import java.util.Scanner;
2
3 public class ArraySum {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the number of elements: ");
9         int n = sc.nextInt();
10
11         int[] arr = new int[n];
12         int sum = 0;
13
14         System.out.println("Enter " + n + " elements:");
15
16         for (int i = 0; i < n; i++) {
17             arr[i] = sc.nextInt();
18             sum = sum + arr[i];
19         }
20     }
21 }
Problems Output Debug Console Terminal Ports
• PS C:\Users\Ayushi Mittal\java programs> javac ArraySum.java
• PS C:\Users\Ayushi Mittal\java programs> java ArraySum
Enter the number of elements: 3
Enter 3 elements:
6
7
8
Sum of array elements = 21
• PS C:\Users\Ayushi Mittal\java programs>
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Java
```

2. Find largest element in an array.

```
import java.util.Scanner;
```

```
public class LargestArray {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number of elements: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        System.out.println("Enter array elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        int largest = arr[0];
```

```
        for (int i = 1; i < n; i++) {
```

```
            if (arr[i] > largest) {
```

```

        largest = arr[i];
    }

}

System.out.println("Largest element = " + largest);

sc.close();
}
}

```

OUTPUT

The screenshot displays the Visual Studio Code interface. On the left, the Explorer panel shows a project named 'java programs' containing various Java files. The main editor window shows the code for 'LargestArray.java'. The code defines a 'LargestArray' class with a 'main' method that takes an array of strings as input. It converts the input to integers, finds the largest element using a for loop, and prints the result. The bottom panel shows the 'Terminal' output, where the program is executed, and the user inputs 6 elements: 7, 8, 9, 0, 6, and 4. The program correctly identifies 9 as the largest element.

```

// LargestArray.java
public class LargestArray {
    public static void main(String[] args) {
        int largest = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }

        System.out.println("Largest element = " + largest);
        sc.close();
    }
}

```

```

PS C:\Users\Ayushi Mittal\java programs> java LargestArray
Enter number of elements: 6
Enter array elements:
7
8
9
0
6
4
Largest element = 9
PS C:\Users\Ayushi Mittal\java programs>

```

3. Search an Element in an Array.

```
import java.util.Scanner;
```

```
public class SearchArray {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number of elements: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        System.out.println("Enter array elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

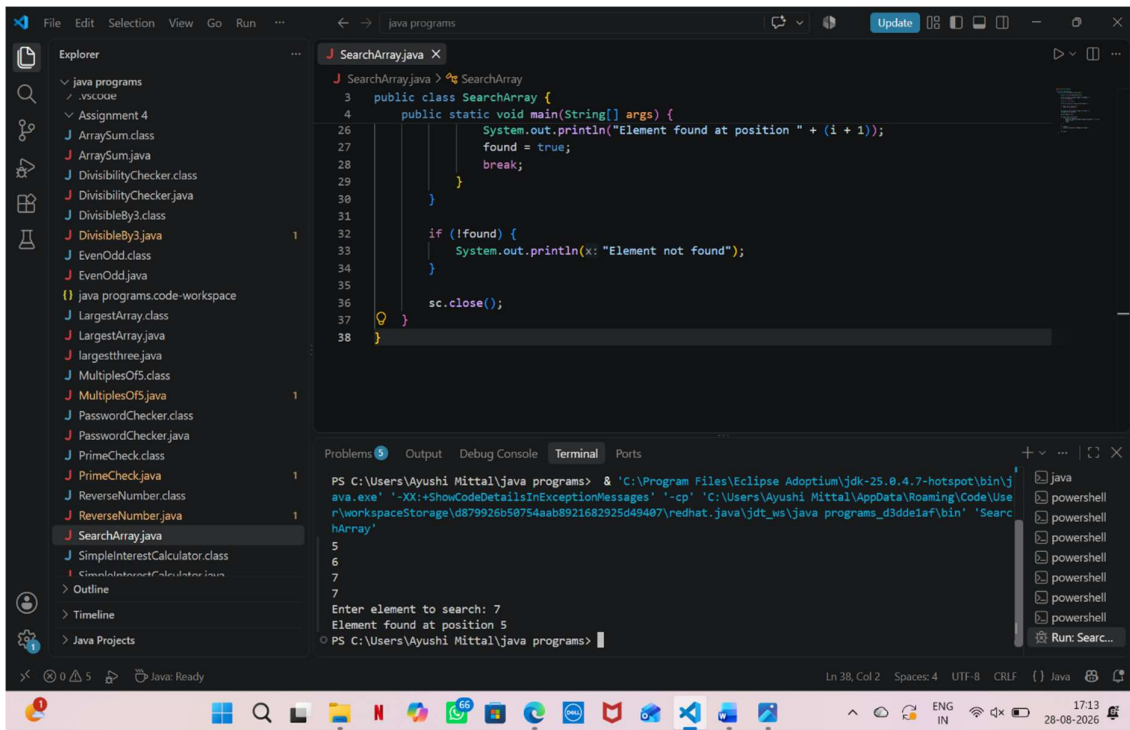
```
        System.out.print("Enter element to search: ");
```

```
        int search = sc.nextInt();
```

```
        boolean found = false;
```

```
for (int i = 0; i < n; i++) {  
    if (arr[i] == search) {  
        System.out.println("Element found at position " + (i + 1));  
        found = true;  
        break;  
    }  
}  
  
if (!found) {  
    System.out.println("Element not found");  
}  
  
sc.close();  
}  
}
```

OUTPUT:



4. Reverse an Array.

```
import java.util.Scanner;

public class ReverseArray {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println("Reversed array:");

        for (int i = n - 1; i >= 0; i--) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

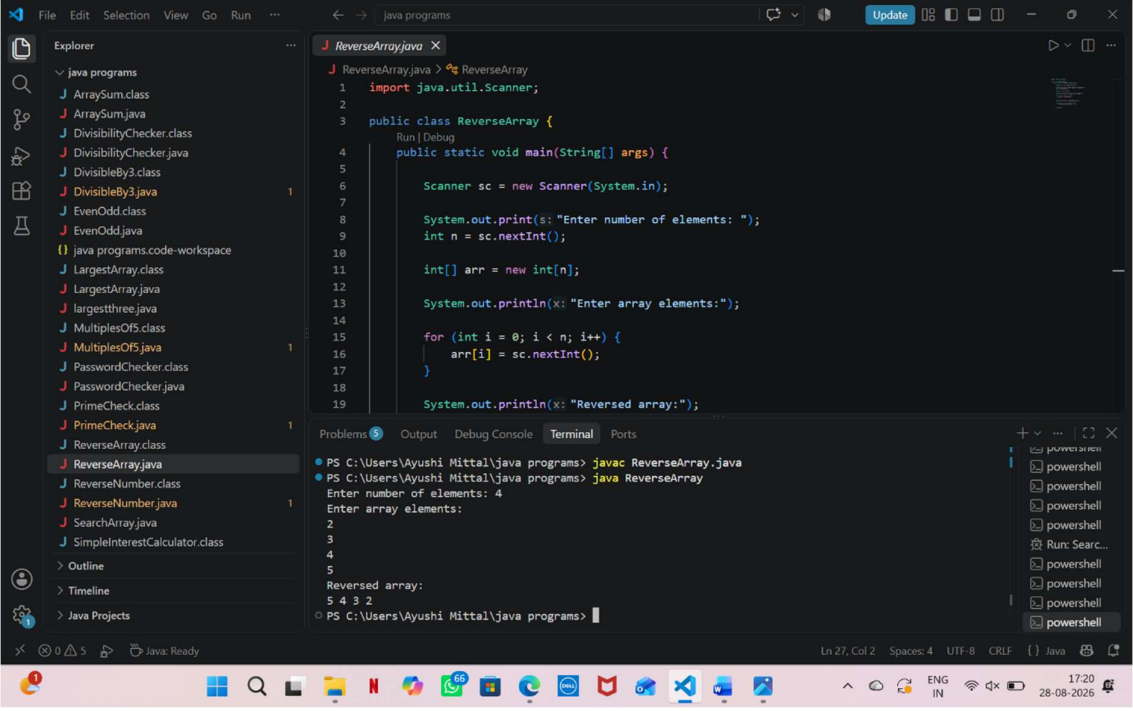
```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT



The screenshot shows an IDE with the following components:

- Explorer:** A list of Java files in the 'java programs' folder, including `ReverseArray.java`, `ReverseNumber.java`, `SearchArray.java`, and others.
- Editor:** The `ReverseArray.java` file is open, showing the following code:

```
1 import java.util.Scanner;
2
3 public class ReverseArray {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print(s: "Enter number of elements: ");
9         int n = sc.nextInt();
10
11         int[] arr = new int[n];
12
13         System.out.println(x: "Enter array elements:");
14
15         for (int i = 0; i < n; i++) {
16             arr[i] = sc.nextInt();
17         }
18
19         System.out.println(x: "Reversed array:");
```
- Terminal:** The output of the program is displayed, showing the prompts and user input:

```
PS C:\Users\Ayushi Mittal\java programs> javac ReverseArray.java
PS C:\Users\Ayushi Mittal\java programs> java ReverseArray
Enter number of elements: 4
Enter array elements:
2
3
4
5
Reversed array:
5 4 3 2
PS C:\Users\Ayushi Mittal\java programs>
```

5. Count Occurrence of an Element in an Array

```
import java.util.Scanner;
```

```
public class CountOccurrence {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number of elements: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        System.out.println("Enter array elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        System.out.print("Enter element to count: ");
```

```
        int element = sc.nextInt();
```

```
        int count = 0;
```

```

for (int i = 0; i < n; i++) {
    if (arr[i] == element) {
        count++;
    }
}

System.out.println("Occurrence of " + element + " = " + count);

sc.close();
}
}

```

OUTPUT:

The screenshot shows an IDE with the following components:

- Explorer:** A list of Java files in the 'java programs' folder, including 'CountOccurrence.java' which is currently selected.
- Editor:** Displays the code for 'CountOccurrence.java'. The code is as follows:


```

public class CountOccurrence {
    public static void main(String[] args) {
        int count = 0;

        for (int i = 0; i < n; i++) {
            if (arr[i] == element) {
                count++;
            }
        }

        System.out.println("Occurrence of " + element + " = " + count);

        sc.close();
    }
}

```
- Terminal:** Shows the execution of the program. The output is:


```

PS C:\Users\Ayushi.Mittal\java programs> javac CountOccurrence.java
PS C:\Users\Ayushi.Mittal\java programs> java CountOccurrence
Enter number of elements: 4
Enter array elements:
5
6
7
8
Enter element to count: 2
Occurrence of 2 = 0
PS C:\Users\Ayushi.Mittal\java programs>

```

